

Department of Computer Science



# FINAL YEAR PROJECT

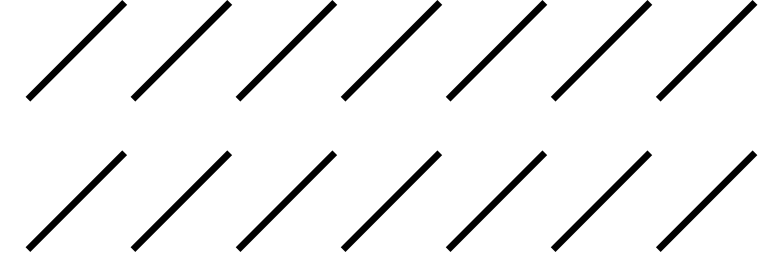
The Complete Beginners Guide

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v3.0

# AGENDA

- 01    Introduction
- 02    Types of Projects
- 03    Blueprint
- 04    Past Projects
- 05    Resources
- 06    Referencing
- 07    Plagiarism
- 08    Supervisors



*section 1*

# **INTRODUCTION**

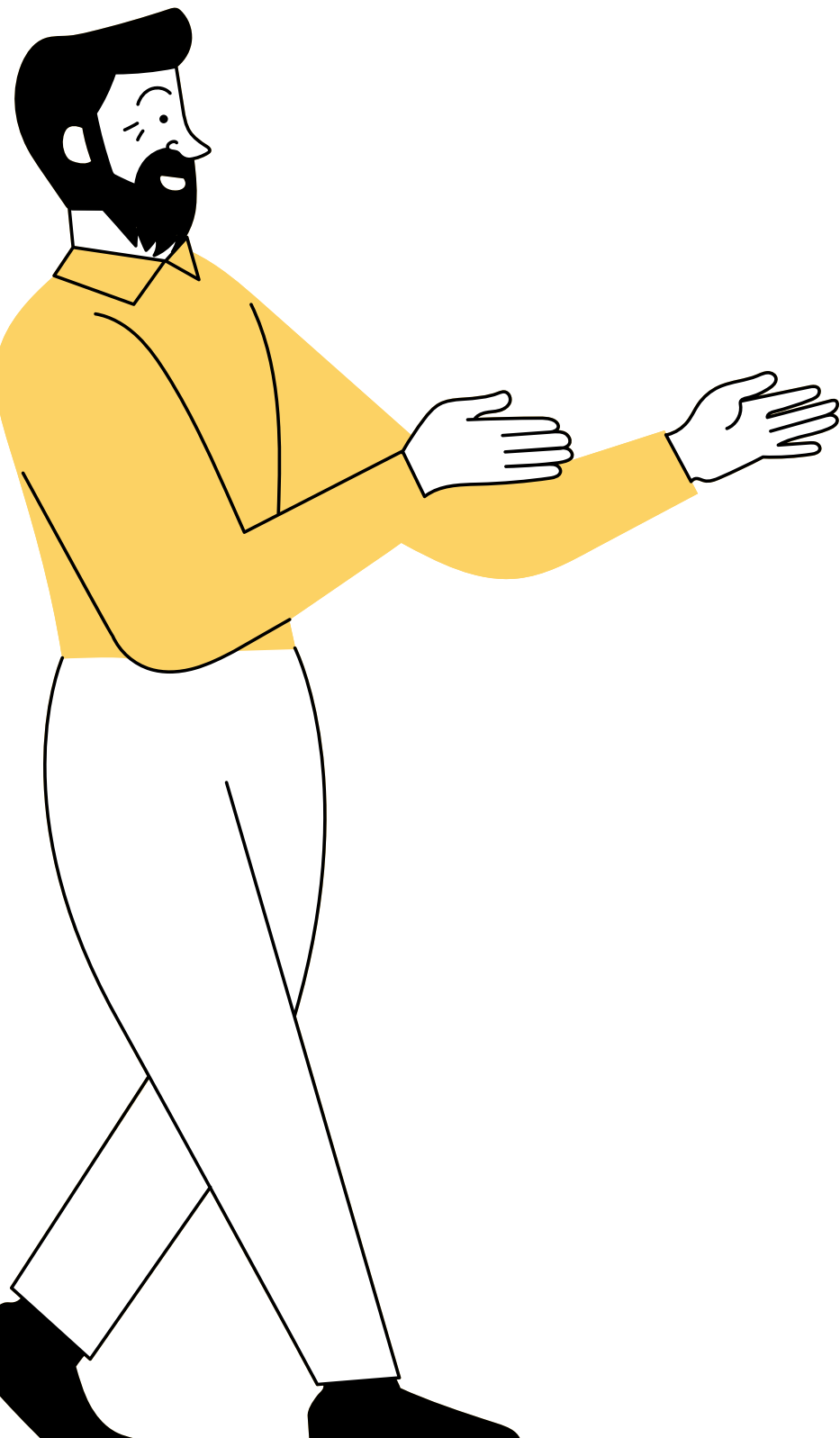
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# 1.1. Timeline

| Phase                  | Y2 - S3 | Y3 - S1 | Y3 - S2 |
|------------------------|---------|---------|---------|
| Project Registration   |         |         |         |
| Module - FYP (40 cred) |         |         |         |
| Module - A (20 cred)   |         |         |         |
| Module - B (20 cred)   |         |         |         |
| Module - C (20 cred)   |         |         |         |
| Module - D (20cred)    |         |         |         |



## 1.2. Why?



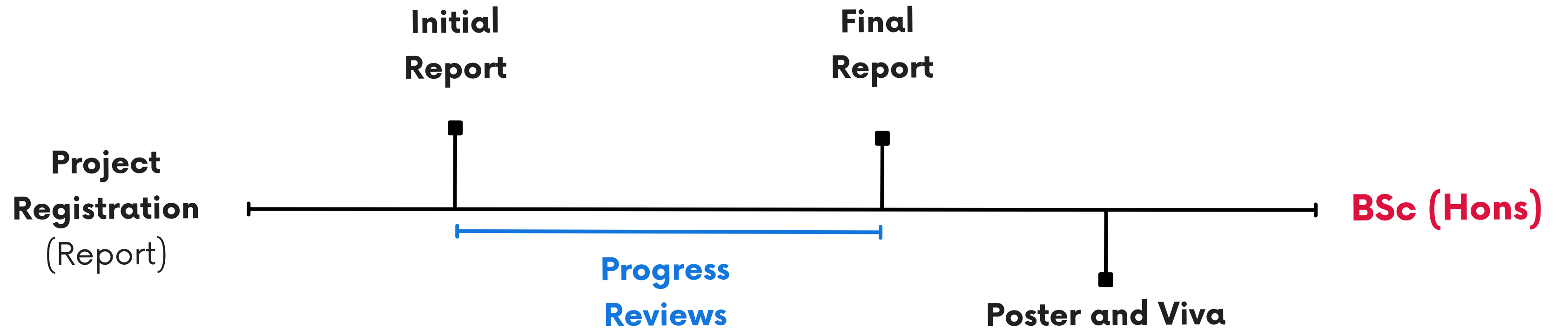
**Degree** An **Honours Degree** develops you as an individual in a field of expertise.

■ **Project** Illustrates to employers your problem-solving, practical, time management, discipline, creativity skills, communication ability

■ **Dissertation** Proves your knowledge and allows you to make a contribution by approving/disapproving new or existing theories

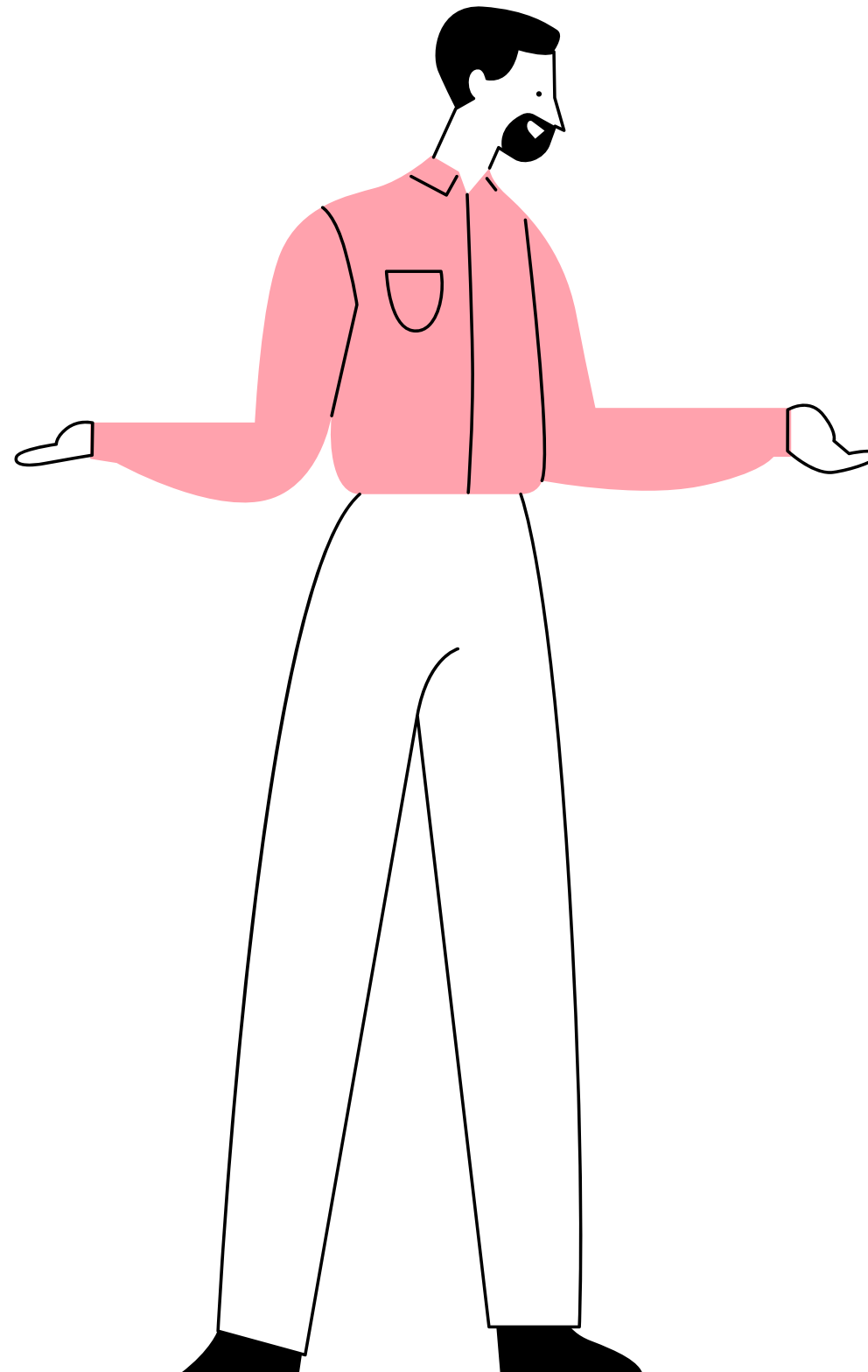
■ **Artefact** Artefact is the technical application (solution) in the form of software/hardware/framework

# 1.3 Journey



## 1.4. Purpose

The final year project offers the chance to delve deeply into a **specialized field** relevant to your course, allowing you to explore a personal **area of interest**.



With guidance from your supervisor, you'll conduct a thorough and research-informed project, culminating in a practical outcome (**Artefact**) accompanied by contextual material (**Dissertation**).

# 1.5. Learning Outcomes

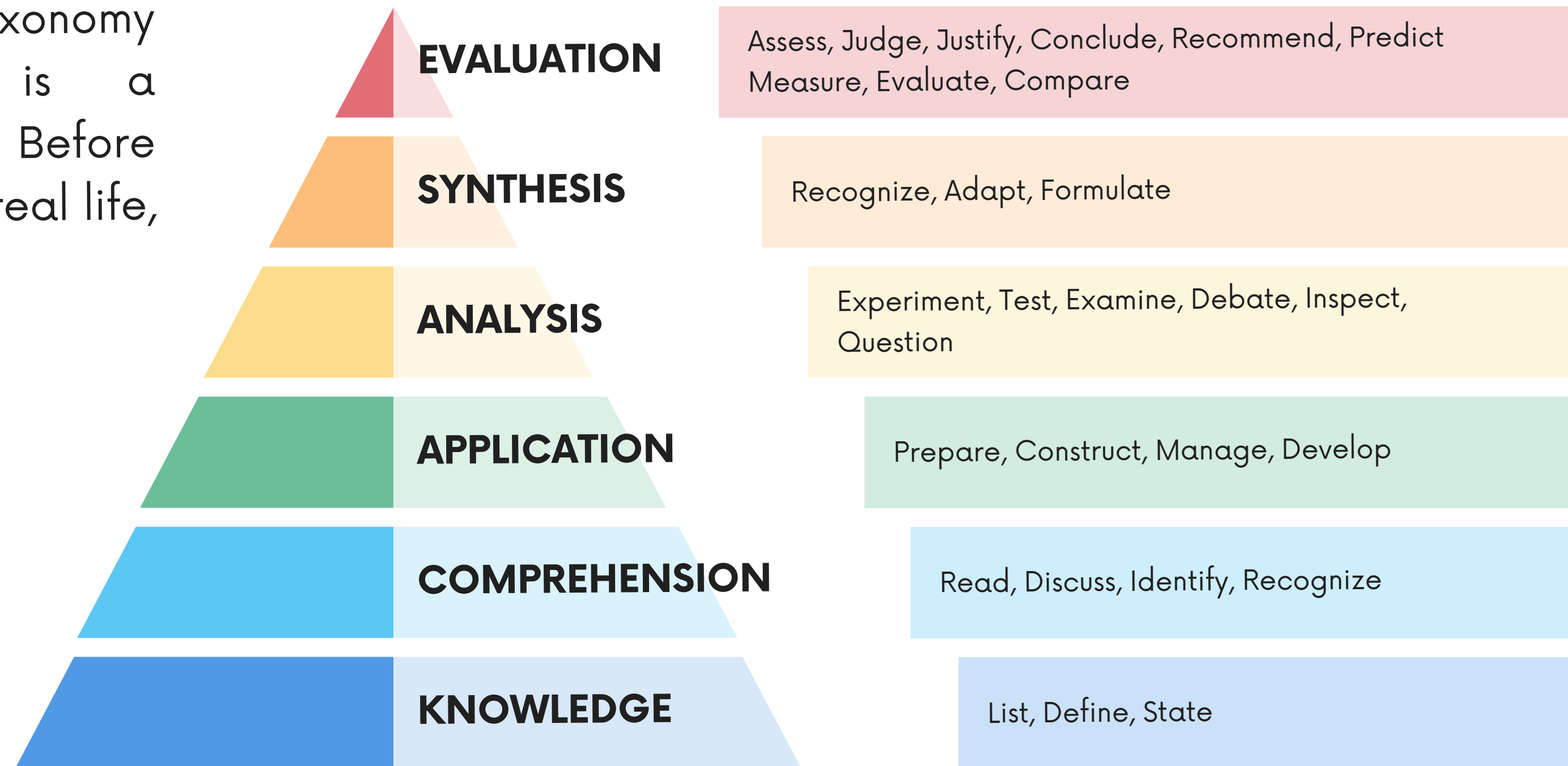
The project module learning outcomes are as follows:

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1. Review the existing academic or professional knowledge base to **scope and substantiate** a discipline-relevant problem.
2. Formulate and manage an iterative project **plan** to address the identified problem, considering feasibility, risks, and ethical aspects.
3. Apply discipline-specific methodologies and techniques to **design** the artefact that addresses the identified problem.
4. **Implement** the designed artefact to address the intended purpose.
5. Critically **evaluate** the artefact's relevance and impact within a suitable societal, industry or scholarly context.
6. Effectively **communicate** the project outcomes to diverse audiences.

# 1.5.1 Blooms Taxonomy

The idea of Bloom's Taxonomy is that learning is a consecutive process. Before applying a concept in real life, we must understand it.





*section 2*

# **TYPES OF PROJECTS**

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## 2.1. Industry-supported

Suggested by industry partners and will be undertaken with some mentoring from industry experts

Applied for in a competitive process



Available on first come first serve basis



## 2.2. Placement-related

Developed during placements for the student to continue within their final year with some input from the original placement organisation

Expected that placement organisations may have a number of other projects that can be treated as industry supported projects



Only viable for placement students





## 2.3. Research-based

Suggested by members of the different research groups in the school and will be supervised and supported by research active staff.



## 2.4. Student/Staff based

Suggested by and supervised by specific academic staff

Discussions between academics and students will lead to the allocation of a project to an individual student



*section 3*

**BLUEPRINT**

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# 3.1. The Idea

Exploring different ideas and concepts within your field. Identify a promising project proposal that reflects your interests and expertise

## **Activity**

- Brainstorm 3-5 project ideas



**Idea X**

## 3.2. Rationale and Benefits

Outline the **value and benefits** to be derived from the project and in particular who will gain from it, apart from you. You should consider how your project could support and impact industry and/or research into your topic

### **Activity**

- Brainstorm 1 rationale for each project idea



**Rationale**



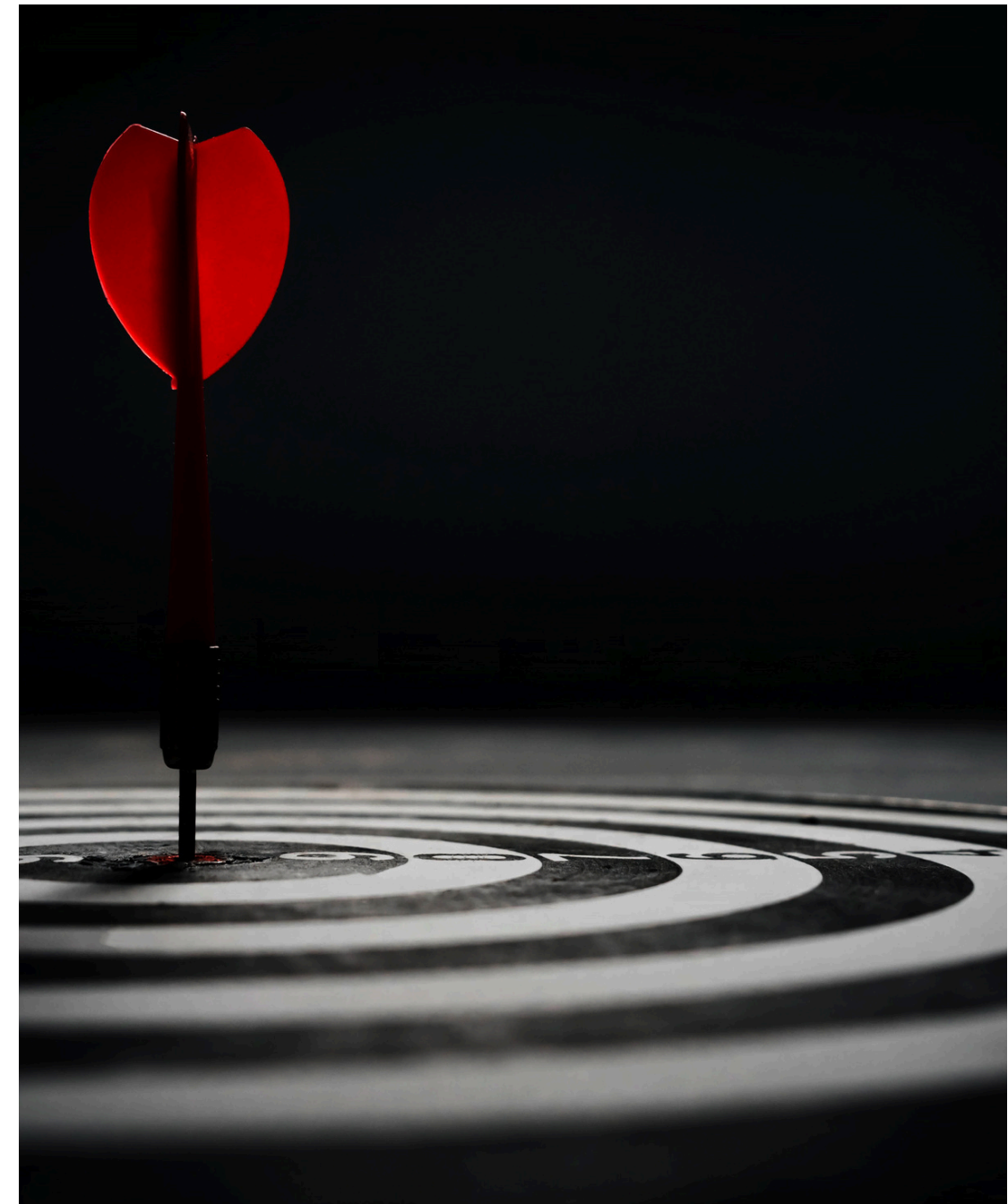
**Idea X**

## 3.3. Aim

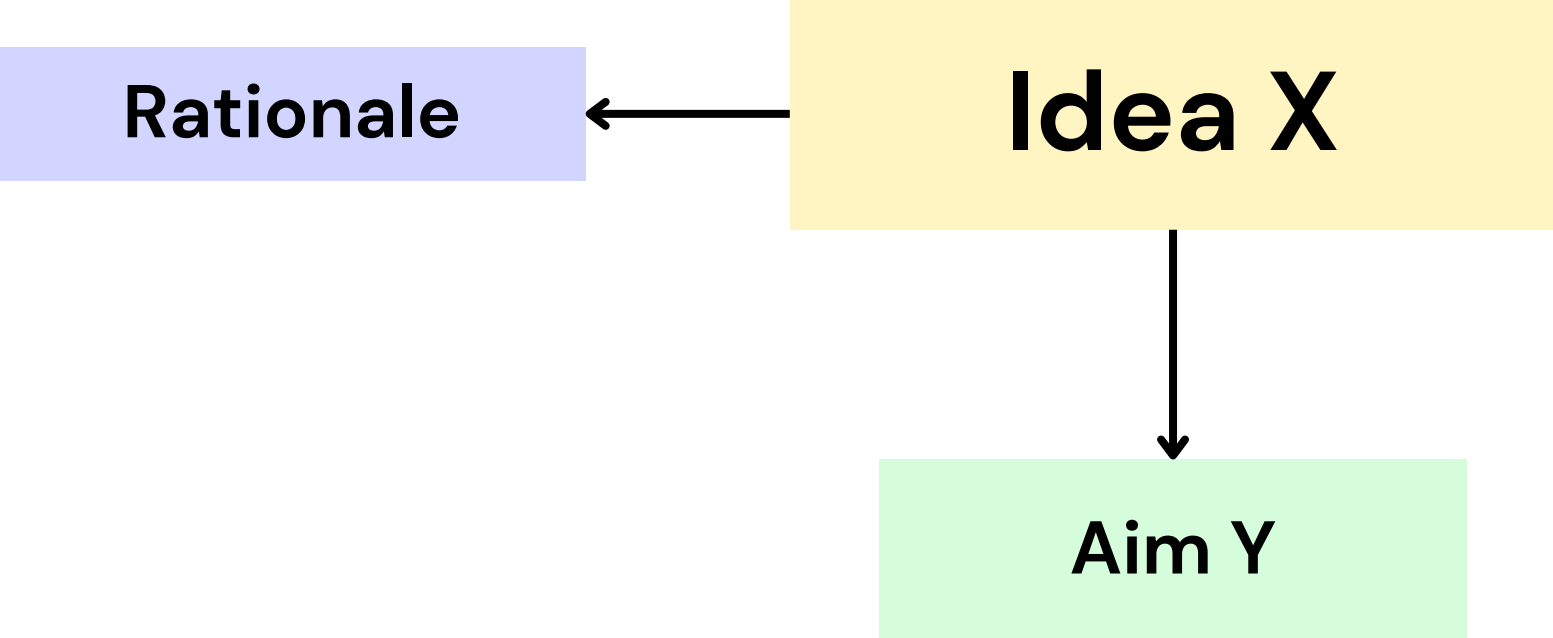
The aim states **what you want to achieve overall** but excludes the method by which you plan to do it. It is a broad and generalised statement of intent

### Activity

- Brainstorm 1 aim for each idea







## 3.4. Objectives

The objectives set out how you are going to achieve your aim. Objectives should be made to be Specific, Measurable, Achievable, Resourced and Time-limited (SMART)

### **Activity**

- Brainstorm 3-5 SMART objectives for each aim

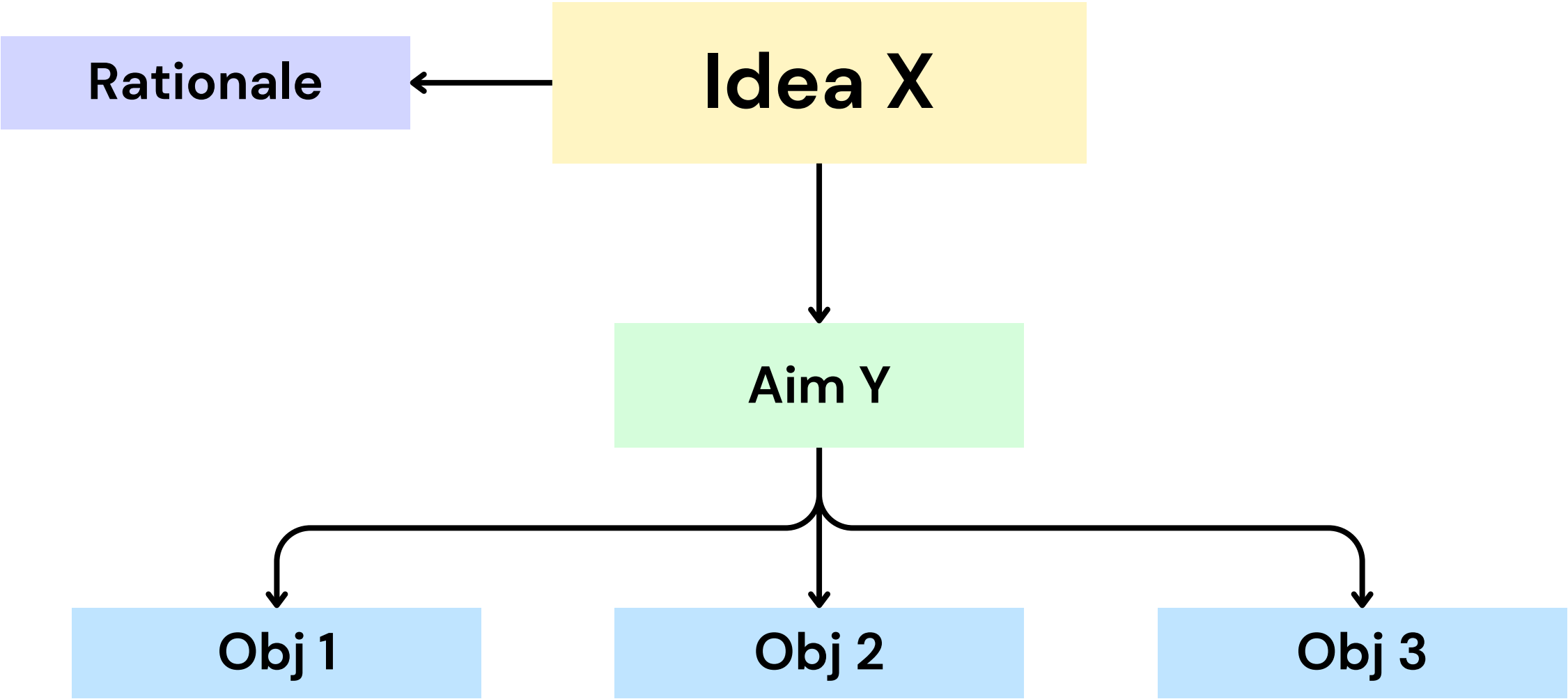


SMART

## 3.4.1 SMART Objectives

SMART objectives are specific, measurable and actionable

|          |                   |       |                                                                                               |
|----------|-------------------|-------|-----------------------------------------------------------------------------------------------|
| <b>S</b> | <b>Specific</b>   | ■ — ■ | Make your goal specific and narrow for more effective planning                                |
| <b>M</b> | <b>Measurable</b> | ■ — ■ | Make sure your goal and progress are measurable                                               |
| <b>A</b> | <b>Achievable</b> | ■ — ■ | Make sure you can reasonably accomplish your goal within a certain time frame                 |
| <b>R</b> | <b>Relevant</b>   | ■ — ■ | Your goal should align with your values and long-term objectives                              |
| <b>T</b> | <b>Time-based</b> | ■ — ■ | Set a realistic but ambitious end date to clarify task prioritization and increase motivation |

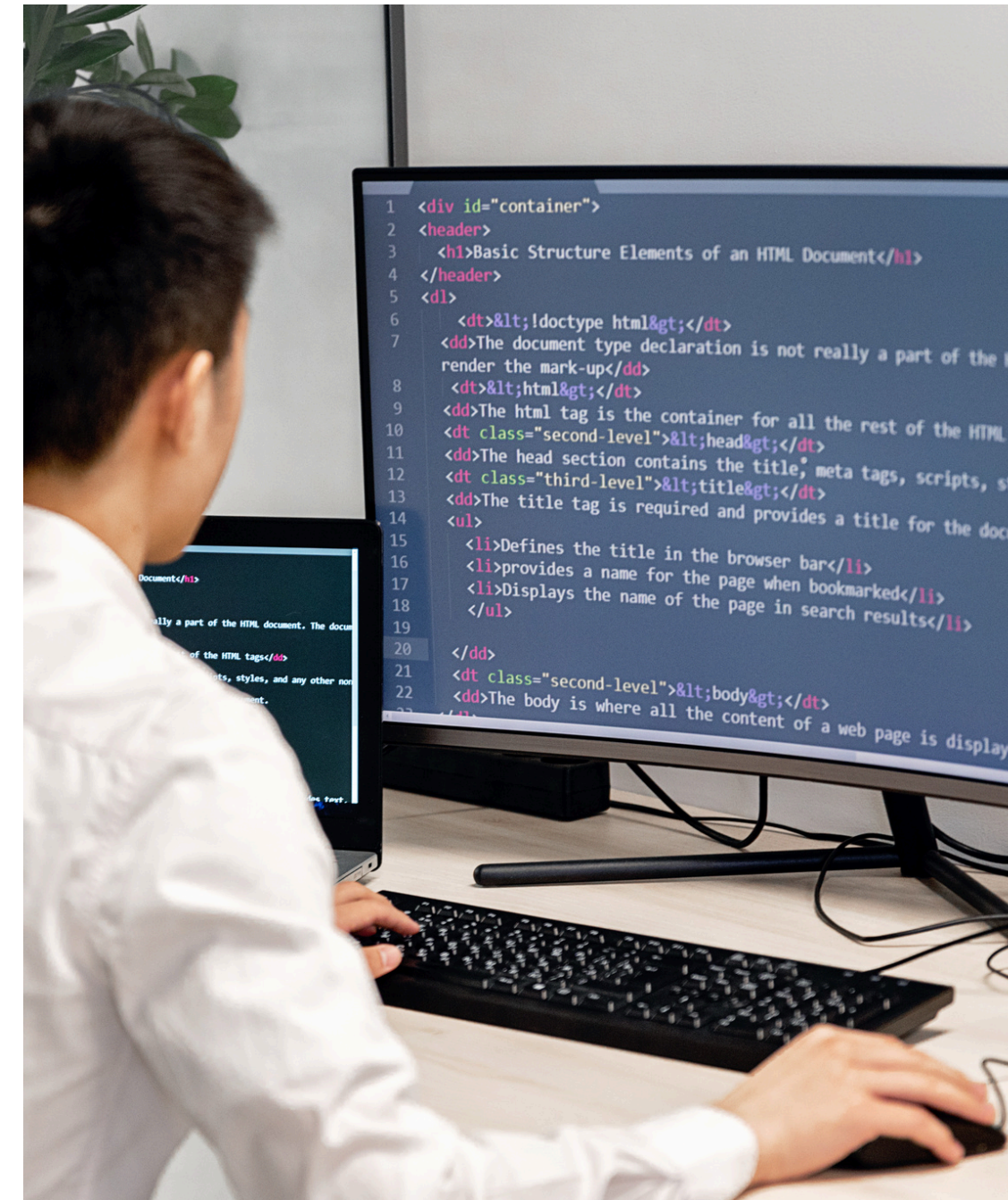


## 3.5 Hardware and Software

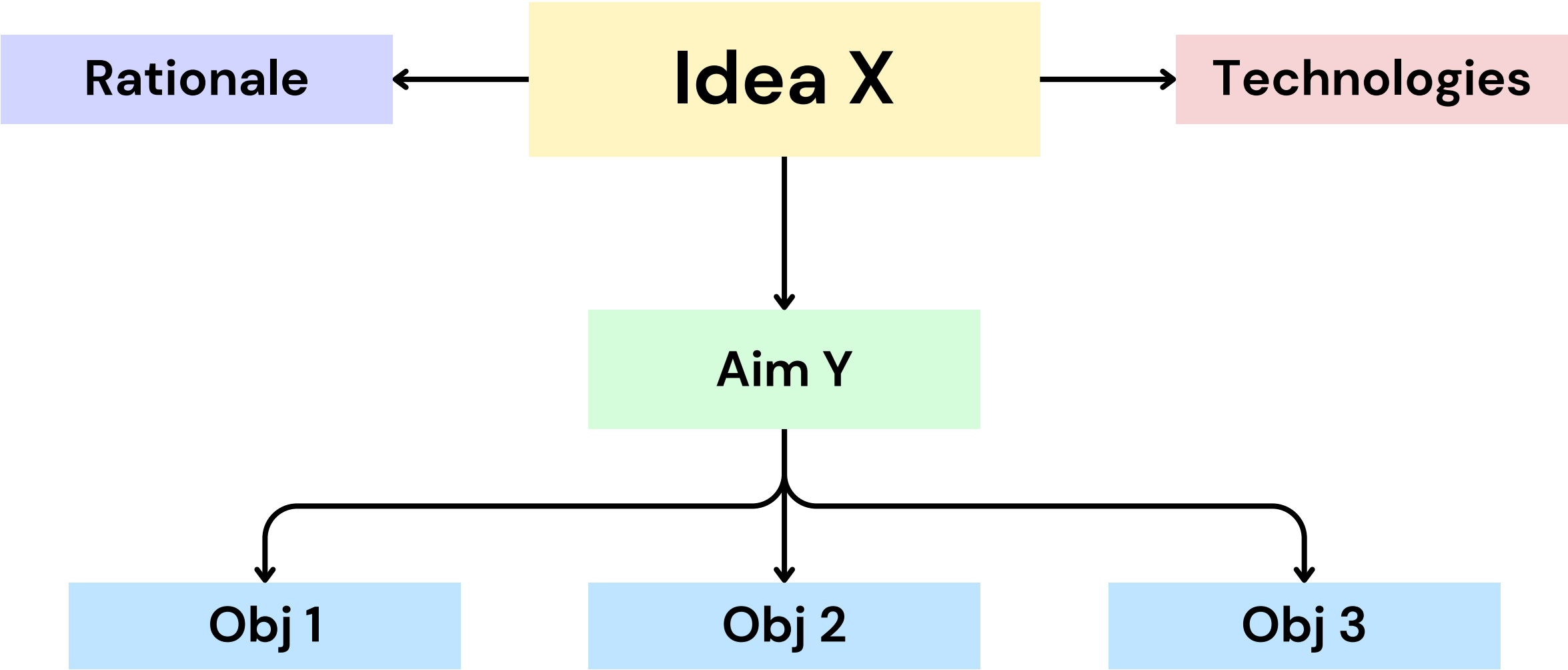
Identify which technologies are relevant and required for you to successfully produce an artefact

### Activity

- Research and identify a list of hardware and/or software you require









*section 4*

# **PAST PROJECTS**

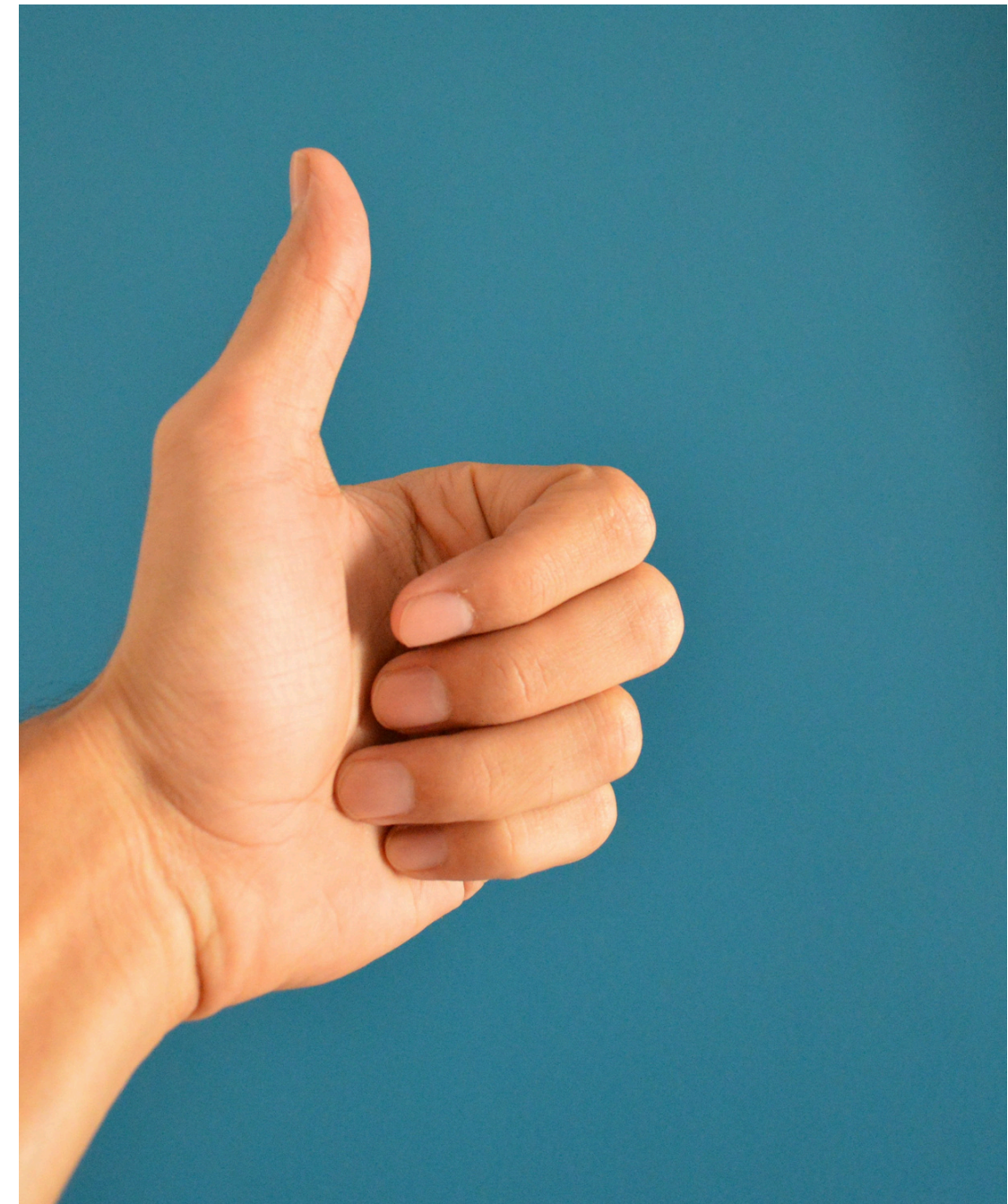
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# 4.1. Critique

Discuss how past projects have influenced your understanding of the final year project and what you see as their key strengths and weaknesses.

## Activity

- Explain the purpose of reviewing previous knowledge, methodologies, and results from the **3** past projects you have chosen and the key points you need to apply to your project



*section 5*

# **RESOURCES**

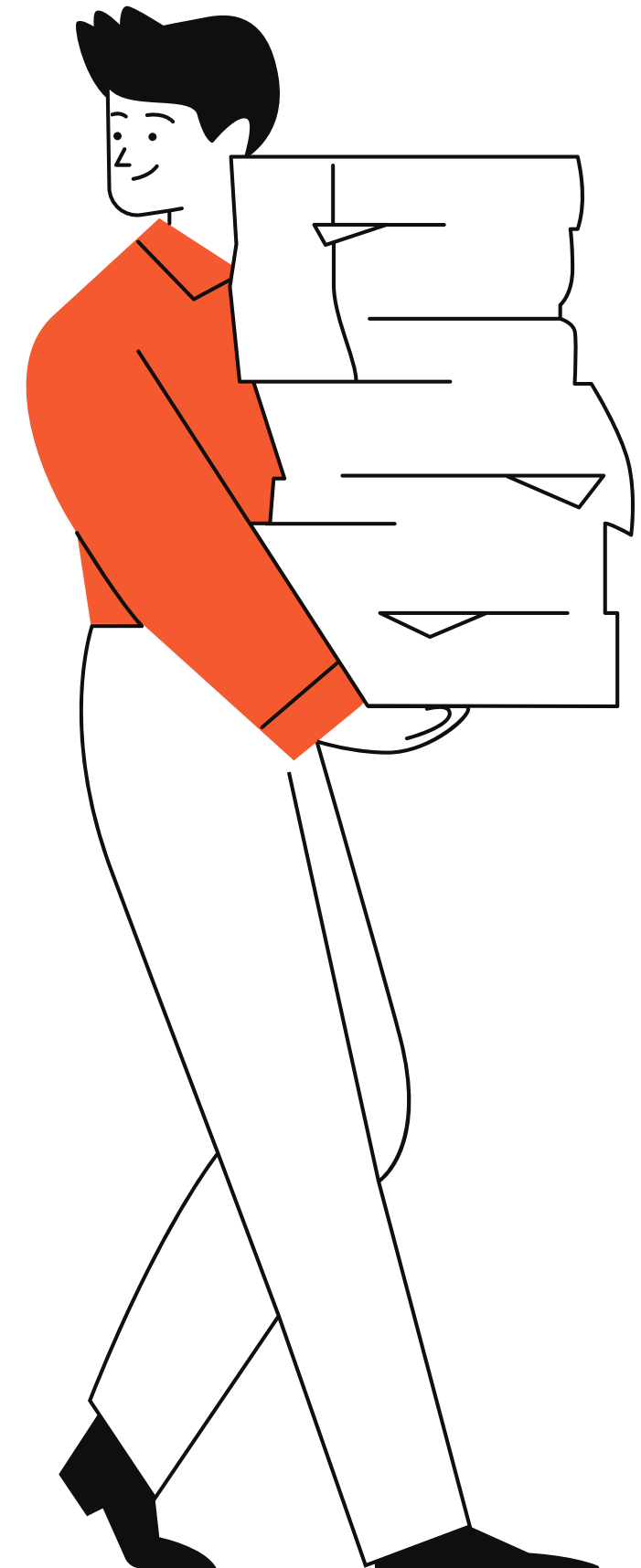
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# 5.1. Literature

Using literature is crucial for providing context, building a theoretical framework, guiding methodology, supporting arguments with evidence, conducting critical analysis, identifying research gaps, and contributing to existing knowledge.

Literature should be:

- **Comprehensive**
- **Appropriate**
- **Current**



## 5.2. Databases

There are a variety of different research databases which can be used



- 1 Scopus ([scopus.com](https://scopus.com))
- 2 Google Scholar ([scholar.google.co.uk](https://scholar.google.co.uk))
- 3 IEEE Xplore ([ieeexplore.ieee.org](https://ieeexplore.ieee.org))
- 4 Science Direct ([sciencedirect.com](https://sciencedirect.com))
- 5 ACM ([dl.acm.org](https://dl.acm.org))
- 6 PubMed ([pubmed.ncbi.nlm.nih.gov](https://pubmed.ncbi.nlm.nih.gov))



## 5.2.1. Boolean Operators

Adding an extra search term  
allows you to refine the search  
further

|                      |                                                                     |
|----------------------|---------------------------------------------------------------------|
| <b>AND</b>           | Finds only those documents that contain all of the terms.           |
| <b>OR</b>            | Finds documents that contain any of the terms.                      |
| <b>AND NOT / NOT</b> | Excludes documents that include the specified term from the search. |



## 5.2.2. Filtering

Adding an extra search term(s) also allows you to further refine your results

### Activity

- Based on your chosen idea use key terms and operators to identify minimum **five** relevant papers

"Artificial Intelligence"



~ 500,00 documents found

"Deep Learning Techniques"



~100,00 documents found

"Convolutional Neural Networks for Image Classification"



~50,00 documents found

"Application of Transfer Learning in Medical Image Analysis"



~500 documents found

## 5.3. Writing Tools

There are a variety of different writing tools which can be used



1

Microsoft Word

2

Overleaf ([overleaf.com](https://overleaf.com))

3

Authorea ([authorea.com](https://authorea.com))

## 5.4. Grammar

Grammar is important in a dissertation because it ensures:

- Clear communication
- Professionalism
- Reader Engagement
- Credibility
- Publication potential

1

Grammarly ([grammarly.com](https://www.grammarly.com))

2

Quilbot ([quillbot.com](https://www.quillbot.com))

3

ProWritingAid ([prowritingaid.com](https://www.prowritingaid.com))



## 5.5. Datasets

Datasets are crucial if your project involves research, model development, hypothesis validation, or expanding the project's scope and applicability

- 1 Kaggle ([kaggle.com/datasets](https://kaggle.com/datasets))
- 2 Google ([datasetsearch.research.google.com](https://datasetsearch.research.google.com))
- 3 Data Gov ([data.gov](https://data.gov))
- 4 Earthdata ([earthdata.nasa.gov](https://earthdata.nasa.gov))
- 5 UCI ([archive.ics.uci.edu/datasets](https://archive.ics.uci.edu/datasets))

*section 6*

# **REFERENCING**

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# 6.1. Why is it important?

Referencing is when you acknowledge that you have used someone else's material or ideas in your own work.

- Allows readers to **source** the information you have used
- Shows the research you have carried out for your work and helps you to **back up your arguments**
- Indicates when you are using someone else's work so that you're not accused of **plagiarism, cheating, or copying.**





## 6.2. Format

Referencing consists of two elements:

- **In-text citations** - In-text citation gives the author(s) and publication date of a source you are referring to.
- **Full references** - Given in **alphabetical order** in the reference list at the end of your work and is not included in the word count



Adhere to **Harvard referencing** guidelines for proper citation and referencing.



# 6.2.1. Authors

| One Author                                                                                                                                                                                                               | Two Authors                                                                                                                                                                                                                      | Three Authors                                                                                                                                                                                                                        | Four Authors                                                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>It has been emphasised that good referencing is an important academic skill (Harris, 2015), 2015).</p> <p>or</p> <p>It has been emphasised by Harris (2015) that good referencing is an important academic skill.</p> | <p>It has been emphasised (Shah and Papadopoulos, 2015) that good referencing is an important academic skill.</p> <p>or</p> <p>Shah and Papadopoulos (2015) emphasised that good referencing is an important academic skill.</p> | <p>It has been emphasised that good referencing is an important academic skill (Wong, Smith and Adebole, 2015).</p> <p>or</p> <p>Wong, Smith and Adebole (2015) emphasised that good referencing is an important academic skill.</p> | <p>It has been emphasised that good referencing is an important academic skill (Wong et al., 2015).</p> <p>or</p> <p>Wong et al. (2015) emphasised that good referencing is an important academic skill.</p> |

# 6.2.2 Example

## In-text Citation

According to recent research in artificial intelligence (AI), deep learning techniques have shown promising results in computer vision tasks ([Ittannavar et al., 2023](#)). Specifically, convolutional neural networks (CNNs) have been widely adopted for image classification, object detection, and semantic segmentation tasks ([Zhou, 2022](#)).

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## Reference List

- 1.M. Zhou, "Research Advanced in Deep Learning Object Detection," 2022 IEEE Conference on Telecommunications, Optics and Computer Science (TOCS), Dalian, China, 2022, pp. 1318-1322, doi: 10.1109/TOCS56154.2022.10016018.
- 2.S. S. Ittannavar, B. P. Khot, M. Mahadevaswamy and R. H. Havaladar, "Deep Learning for Computer Vision: Recent Breakthroughs and Emerging Trends," 2023 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES), Chennai, India, 2023, pp. 1-5, doi: 10.1109/ICSES60034.2023.10465486.

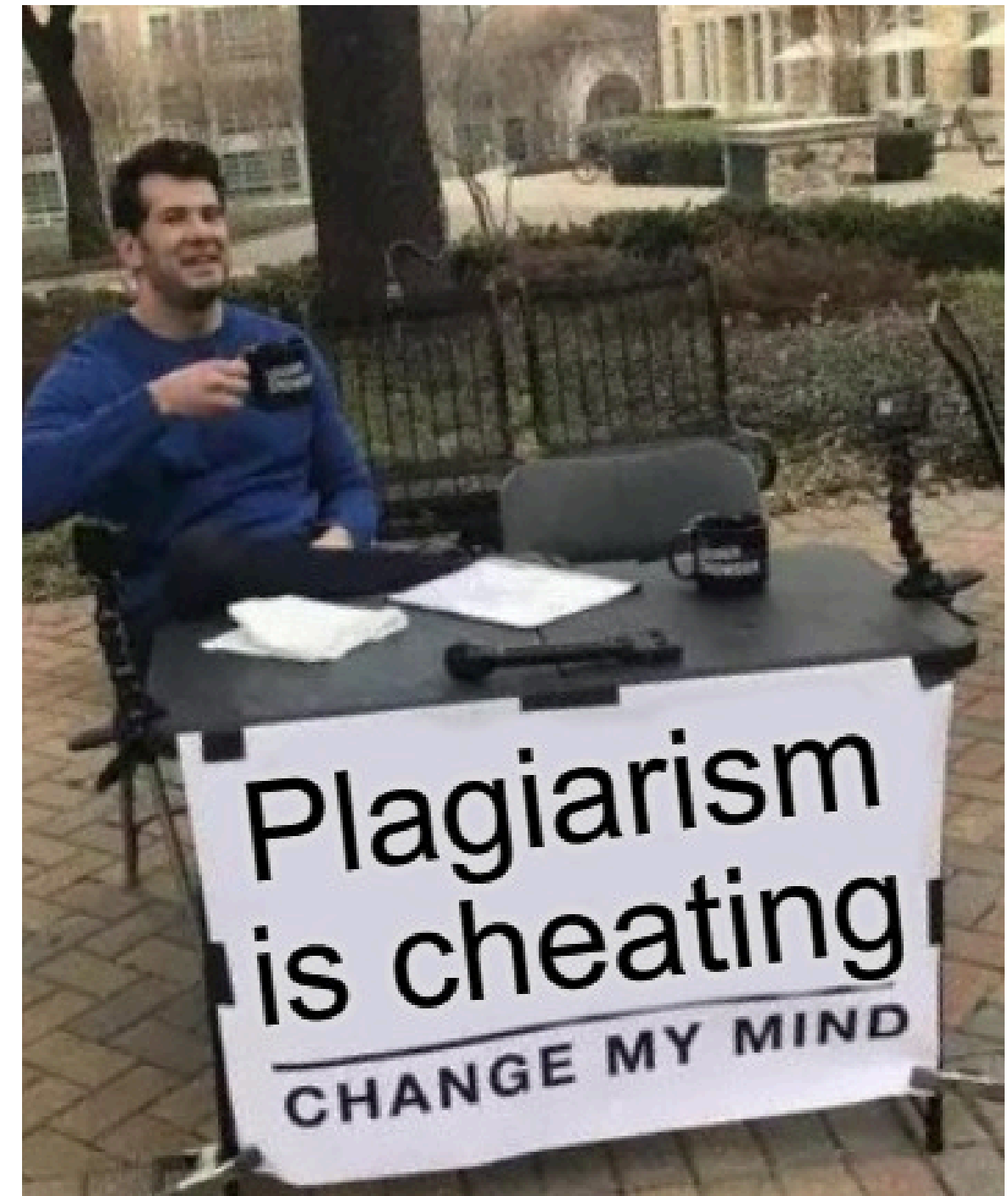
*section 7*

# **PLAGIARISM**

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# 7.1 What is it?

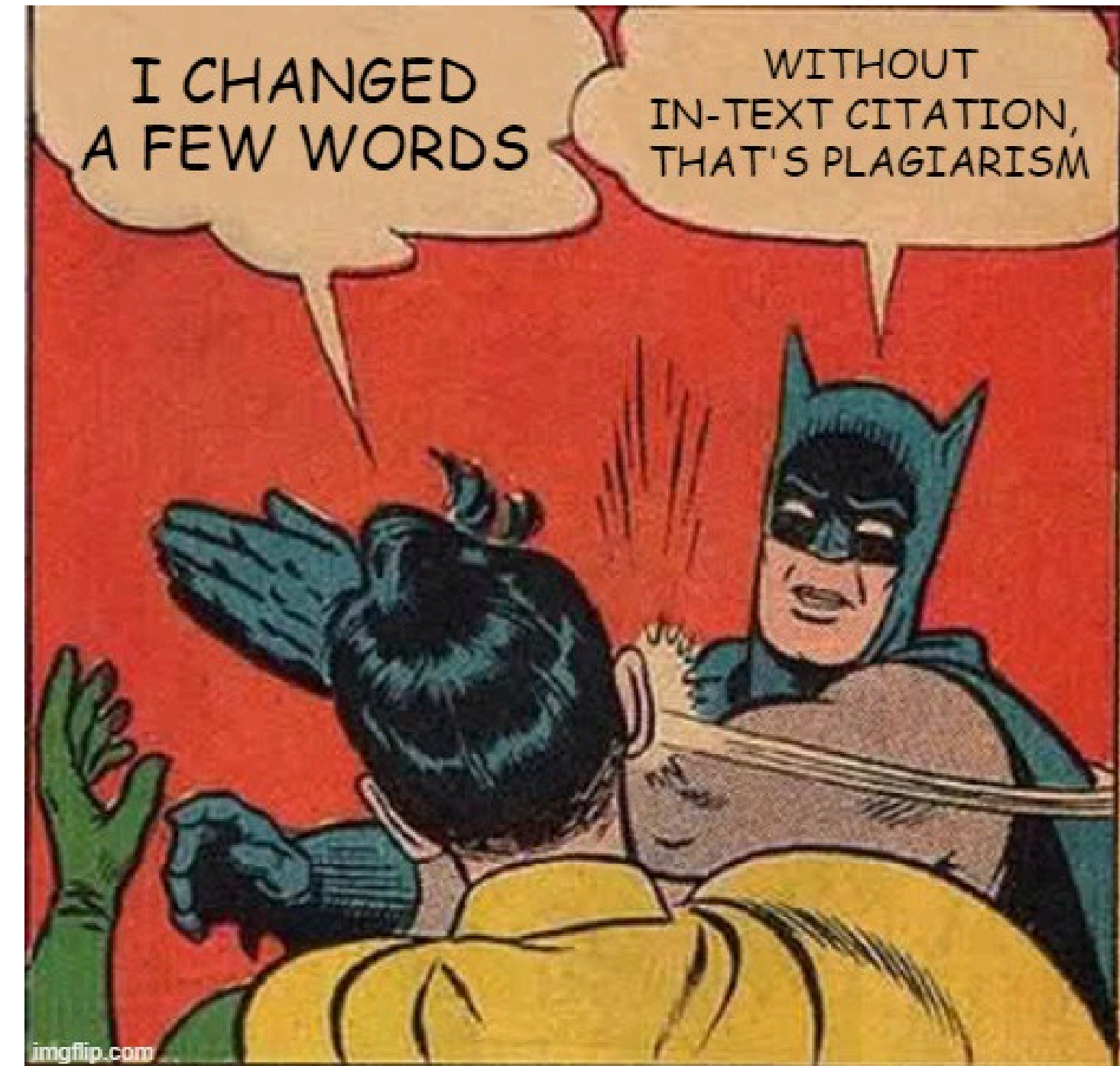
The university defines plagiarism as: "...submitting any item of assessment which contains work produced by someone else in a way which makes it look as though it is your own work." (BCU, 2023).





## 7.2 What **NOT** to do

-  Copying any kind of published material and presenting it as your own
-  Poor paraphrasing or summarising
-  Not referencing your work properly
-  Copying the work of another student
-  Submitting work that you submitted previously – self-plagiarism
-  AI misuse





# **AI Tools at BCU: Student Guidelines**



# **AI tools Decision Tree**



*section 8*

# **SUPERVISORS**

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# 8.1 Supervisory Team



## 1st Supervisor

Supports any technical challenges or issues you face; this is **your main supervisor**

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Regular group meetings.

1-1 meetings for more detailed specific issues.  
involved in all stages of the project and assessment.



## 2nd Supervisor

Assists through more general academic guidance and support throughout the entire project

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May not be a subject specialist.  
Involved in assessment.